

Q-Folio: one-page business brief

Qiskit Fall Fest 2026 · Track I2 · Achintya Akella

Customer. Indian wealth-tech and robo-advisory platforms that sell curated, equal-weight "model baskets" (themed portfolios of a few stocks) to retail investors. Each platform maintains hundreds of baskets across risk profiles and rebalances them periodically.

The problem and its cost. Choosing *which k stocks* go into a basket is a combinatorial optimisation. The number of candidate baskets grows as $C(n,k)$: 20 options for 3-of-6, ~185 000 for 10-of-20, and astronomically more for real universes. Platforms rely on heuristics and analyst judgement, and a poorly chosen basket shows up directly as lower risk-adjusted return for clients. The market is large and still growing: India had **23.4 crore (234.4 million) demat accounts in July 2026**, up 1.25% in one month ([360 ONE Capital data via ANI](#)).

What we built. A Qiskit pipeline that turns a basket-selection problem (expected returns, covariance, basket size k , client risk aversion q) into a quantum circuit. We use a **constraint-preserving QAOA** (Dicke initial state + XY mixer): the circuit only ever proposes baskets with exactly k stocks. It runs on simulators, on IBM noise models and on a real IBM quantum computer, and it is checked against exact and heuristic classical solvers.

Result (3-of-6 baskets, 2023–2025 NSE prices, $p = 2$, 8192 shots):

- **Validity.** 100% of XY-QAOA samples were valid baskets. Standard penalty-QAOA produced only 75% valid baskets and found the optimum at **chance level** ($P(\text{optimal})$ 0.04 vs 0.05 random).
- **Robustness, 24 instances.** XY-QAOA sampled the optimal basket with $P = 0.34 \pm 0.16$ vs **0.05** for a random valid basket, comparable to simulated annealing on the same objective (0.36 ± 0.08). Its most frequent answer was the optimum on 75% of instances.
- **The real NSE basket (6 large caps).** The best and second-best baskets differ by only 1.4% of the cost range, so we report $P(\text{top-2})$ and approximation ratio here: XY-QAOA $P(\text{top-2})$ **0.37** vs 0.10 random, approximation ratio **0.84** vs 0.52 random.
- **Real IBM hardware** (`ibm_kingston`, 4 stocks, $k = 2$, job `db3b1bimb58s7387e0jg`): $P(\text{optimal})$ **0.51 raw** → **0.65 after symmetry post-selection** vs 0.17 random, and the top answer was the true optimum.

Limits (read this).

- At this size a classical computer finds the exact answer in under a millisecond (brute force, greedy). **There is no quantum advantage today**, and we do not claim one.
- Today's hardware noise limits us to ~4–6 assets and shallow circuits (22% of raw hardware shots were invalid baskets).
- The model is simplified: equal weights, historical estimates, no costs or sector rules.

Why it still matters / next steps. The project shows that **encoding business constraints directly in the circuit** is necessary for quantum optimisation to work at all, and that it also gives built-in error detection. Next steps: sector and turnover constraints as further symmetries, parameter transfer across baskets, and benchmarking against commercial MIQP solvers as hardware matures.

Links: [Repo](#) · [Report](#) · [Live demo](#) · Video (YouTube URL)

Public market data only. Not investment advice. All numbers from `results/benchmark.json` and `results/hardware/db3b1bimb58s7387e0jg.json`.